학 사 학 위 논 문
Bachelor's Thesis

중성원자 배열에서의 조합최적화: 문제 인코딩, 잡음 완화, 그리고 근사해 성능 평가

Combinatorial Optimization on Neutral Atom Arrays: Problem Encoding, Noise Mitigation, and Approximate-Solution Performance Evaluation

2026

정 준 우 (鄭 俊 雨 Jung, Jun-Woo)

한 국 과 학 기 술 원

Korea Advanced Institute of Science and Technology

학 사 학 위 논 문

중성원자 배열에서의 조합최적화: 문제 인코딩, 잡음 완화, 그리고 근사해 성능 평가

2026

정 준 우

한 국 과 학 기 술 원

물리학과

중성원자 배열에서의 조합최적화: 문제 인코딩, 잡음 완화, 그리고 근사해 성능 평가

정 준 우

위 논문은 한국과학기술원 학사학위논문으로
학위논문 심사위원회의 심사를 통과하였음

2026년 6월 20일

심사위원장 안 재 욱 (인)

심 사 위 원 최 재 윤 (인)

심 사 위 원 정 하 웅 (인)

Combinatorial Optimization on Neutral Atom Arrays: Problem Encoding, Noise Mitigation, and Approximate-Solution Performance Evaluation

Jun-Woo Jung

Advisor: Jaewook Ahn

A thesis submitted to the faculty of
Korea Advanced Institute of Science and Technology in
partial fulfillment of the requirements for the degree of
Bachelor of Science in Physics

Daejeon, Korea
June 20, 2026

Approved by

Jaewook Ahn
Professor of Physics

The study was conducted in accordance with Code of Research Ethics[1].

[1] Declaration of Ethical Conduct in Research: I, as a graduate student of Korea Advanced Institute of Science and Technology, hereby declare that I have not committed any act that may damage the credibility of my research. This includes, but is not limited to, falsification, thesis written by someone else, distortion of research findings, and plagiarism. I confirm that my thesis contains honest conclusions based on my own careful research under the guidance of my advisor.

BPH

정준우. 중성원자 배열에서의 조합최적화: 문제 인코딩, 잡음 완화, 그리고 근사해 성능 평가. 물리학과 . 2026년. 20+iv 쪽. 지도교수: 안재욱. (한글 논문)

Jun-Woo Jung. Combinatorial Optimization on Neutral Atom Arrays: Problem Encoding, Noise Mitigation, and Approximate-Solution Performance Evaluation. Department of Physics . 2026. 20+iv pages. Advisor: Jaewook Ahn. (Text in Korean)

초 록

중성원자 리드버그 배열은 리드버그 봉쇄를 이용해 그래프의 독립집합 구조를 구현하는 플랫폼이다. 본 보고서는 저자가 참여한 세 연구를 문제 인코딩, 잡음 있는 측정값의 추론, 성능 평가라는 흐름으로 종합한다. 첫째, 전역 제어 조건에서 리드버그 원자 그래프로 이차 비제약 이진 최적화 문제를 구현하는 방법을 정리한다. 둘째, 측정 잡음을 해밍 껍질 구조로 모델링하고 결정론적 오류 완화의 후처리 비용을 이진 엔트로피로 기술한다. 셋째, 후처리된 출력 분포를 최대 독립집합으로부터의 껍질 지수로 표현하고, 목표 근사비에 따른 필요 샷 수를 정의한다. 이를 통해 하드웨어 인코딩, 고전 후처리, 성능 지표를 함께 고려해야 함을 보인다.

핵 심 낱 말 중성원자 배열, 리드버그 원자, 조합최적화, 최대 독립집합, 이차 비제약 이진 최적화, 결정론적 오류 완화, 근사해 샷 복잡도

Abstract

Neutral-atom Rydberg arrays provide a programmable platform for combinatorial optimization by encoding graph constraints through the Rydberg blockade. This graduation research report synthesizes three studies by the author into a unified workflow: problem encoding, inference from noisy measurements, and performance evaluation. First, it reviews how Rydberg-atom graphs can implement quadratic unconstrained binary optimization problems under global control. Second, it describes deterministic error mitigation for noisy measurement bitstrings using a Hamming-shell noise model, where the classical postprocessing cost is governed by binary entropy. Third, it introduces a shell-based description of postprocessed outputs and defines a shots-to-approximate-solution metric for a target approximation ratio. Taken together, these studies show that neutral-atom quantum optimization cannot be evaluated from hardware dynamics alone. A fair assessment must account for the physical encoding overhead, the cost of classical inference from noisy samples, and the operational target accuracy of the optimization task.

Keywords neutral atom arrays, Rydberg atoms, combinatorial optimization, maximum independent set, quadratic unconstrained binary optimization, deterministic error mitigation, shots-to-approximate-solution

차 례

표 차례

그림 차례

제 1 장 Introduction

1.1 Research Background

Combinatorial optimization problems include a class of problems in which the number of possible configurations grows exponentially with system size. Representative examples include the maximum independent set (MIS), maximum cut (MaxCut), satisfiability (SAT), and quadratic unconstrained binary optimization (QUBO). These problems arise across diverse fields such as logistics, scheduling, graph analysis, statistical physics, and machine learning, and in many cases they are classically hard to solve.

Recently, neutral-atom Rydberg arrays have attracted attention as a promising platform for physically implementing combinatorial optimization problems. Optical tweezers allow atoms to be positioned individually, and atoms excited to Rydberg states experience strong van der Waals interactions. This interaction suppresses the simultaneous excitation of two atoms within a certain distance to the Rydberg state, a phenomenon known as the Rydberg blockade. By mapping graph vertices to atoms and edges to blockade interactions, the set of atoms that can be simultaneously excited corresponds naturally to an independent set of the graph.

From this perspective, a neutral-atom array is not merely a quantum simulator but a physical computing device that turns a combinatorial optimization problem into a ground-state search problem of a Hamiltonian. Evaluating an actual quantum optimization experiment, however, requires questions at three levels. First, which combinatorial optimization problems can be encoded onto a Rydberg array, and how? Second, from noisy experimental measurements, what classical postprocessing can recover valid solutions? Third, by what metric should the quality of the obtained solutions and the required number of shots be evaluated?

This report restructures three studies in which the author participated according to these three questions. The first study presents a method for implementing QUBO problems using Rydberg-atom graphs [1]. The second study analyzes the classical computational cost required to perform deterministic error mitigation (DEM) on noisy measured bitstrings, in terms of Hamming shells and binary entropy [2]. The third study defines and analyzes the shots-to-approximate-solution (STS) metric—the number of shots needed to reach a target approximation ratio—based on the shell distribution of postprocessed quantum annealing outputs [3].

1.2 Research Goals and Report Structure

The goal of this report is to describe the three studies not as a list of independent results but as a single unified research workflow. The central perspective is as follows.

> Neutral-atom-based combinatorial optimization consists of a stage that encodes the problem into a physical Hamiltonian, a stage that postprocesses noisy measurement outcomes to recover valid candidate solutions, and a stage that evaluates the success probability and shot complexity for a target accuracy.

Chapter 2 introduces the basic physics of the Rydberg blockade and MIS encoding. Chapter 3 explains the QUBO encoding method using Rydberg-atom graphs. Chapter 4 describes measurement noise, the Hamming-shell structure, and the cost model of DEM. Chapter 5 organizes the postprocessed output distribution into a shell model and introduces the STS metric. Chapter 6 presents an integrated interpretation connecting the three studies, and Chapter 7 discusses conclusions and future work.

Problem encoding
Rydberg graph → Quantum annealing
Measurement bitstrings → Noise inference
Error mitigation → Performance evaluation
Shots to solution

그림 1.1: Conceptual flow of the three studies integrated in this report. The Rydberg-atom array encodes a combinatorial optimization problem as a physical graph, and experimental measurement bitstrings become candidate solutions through noise inference and postprocessing. Performance is finally evaluated using the number of shots required for a target approximation ratio.

제 2 장 Rydberg Atom Arrays and Maximum Independent Set Encoding

2.1 Rydberg Graph Hamiltonian

Let an atom array represent a graph $G = (V, E)$. Each vertex $i \in V$ corresponds to one atom, and the two atomic states $|0\rangle$ and $|1\rangle$ denote the ground state and the Rydberg-excited state, respectively. The standard Hamiltonian of a Rydberg-atom graph can be written as

$$\hat{H}_G = \frac{\Omega}{2} \sum_{i \in V} \hat{\sigma}_i^x - \Delta \sum_{i \in V} \hat{n}_i + \sum_{(i,j) \in E} U_{ij} \hat{n}_i \hat{n}_j, \tag{2.1}$$

where Ω is the Rabi frequency, Δ the detuning, and $\hat{n}_i = |1\rangle_i \langle 1|$ the Rydberg excitation number operator. The interaction is approximately $U_{ij} = C_6/|\mathbf{r}_i - \mathbf{r}_j|^6$, and when two atoms lie within the blockade radius, the state in which they are simultaneously excited carries a large energy cost.

Consider the limit $0 < \Delta < U$ and $\Omega \to 0$; the two terms of the Hamiltonian compete. The $-\Delta \sum_i \hat{n}_i$ term favors exciting as many atoms as possible, while the $U_{ij} \hat{n}_i \hat{n}_j$ term suppresses the simultaneous excitation of two edge-connected atoms. The low-energy states therefore correspond to selecting as many vertices as possible while never simultaneously selecting two adjacent vertices. This is precisely the maximum independent set problem.

2.2 Bitstrings, Independent Sets, and Postprocessing

In an experiment, a single shot produces a bitstring of the form $z \in \{0, 1\}^N$. Here $z_i = 1$ indicates that vertex i was selected. Ideally, since the Rydberg blockade completely suppresses the simultaneous excitation of two adjacent vertices, the measured bitstring should itself be an independent set. In real experiments, however, finite Rabi driving, control errors, atom loss, and measurement noise can cause blockade violations or defects. The experimental bitstring is therefore naturally regarded not as the final solution but as the input to postprocessing.

The three studies treated in this report all share this fact. In the QUBO encoding study, the graph structure itself is designed to implement the desired cost function. In the DEM study, the Hamming space around a noisy measured bitstring is searched to recover a valid solution. In the STS study, the same postprocessing is applied, and which shell near the optimum the output lands in is analyzed statistically.

제 3 장 Quadratic Unconstrained Binary Optimization Encoding Using Rydberg-Atom Graphs

Underlying study of this chapter: A. Byun, J. Jung, K. Kim, M. Kim, S. Jeong, H. Jeong, and J. Ahn, "Rydberg-Atom Graphs for Quadratic Unconstrained Binary Optimization Problems," *Advanced Quantum Technologies*, 2300398 (2024).

Author's contribution: The author (Junwoo Jung) was responsible for the theoretical contribution of this study. Specifically, the author derived the ground-state equivalence conditions for each of the four basic gadgets (data qubit, offset qubit, and even- and odd-atom quantum wires) and contributed to constructing the theoretical framework that combines them to represent a general integer-coefficient QUBO Hamiltonian as a Rydberg-atom graph. The experimental design and data acquisition were carried out by the co-authors, including lead author Byun.

3.1 QUBO Problems and the Need for a Rydberg Implementation

The quadratic unconstrained binary optimization (QUBO) problem is the problem of minimizing the following cost function over binary variables $x_i \in \{0, 1\}$:

$$f(\mathbf{x}) = \sum_{i=1}^{N} Q_{ii} x_i + \sum_{i<j} Q_{ij} x_i x_j. \tag{3.1}$$

QUBO is a universal form capable of expressing a wide range of combinatorial optimization problems. Existing annealing devices and Ising machines also frequently use problems in QUBO or Ising form. To implement QUBO on a neutral-atom platform, the signs and magnitudes of the linear and quadratic terms must be expressed through the geometric structure of a Rydberg-atom graph.

The core of the first study in which the author participated is the construction of Rydberg-atom-graph gadgets that can implement general integer-coefficient QUBO using only global laser control. Real coefficients can be approximated by rationals, and since multiplying the entire cost function by a positive constant does not change the optimum, the problem can in principle be converted to integer-coefficient QUBO.

3.2 Ground-State Equivalence

An important concept in the Rydberg-atom-graph approach is ground-state equivalence. Two Hamiltonians need not be identical at all energy levels; if their ground-state sets coincide, they can be regarded as equivalent for the purpose of obtaining the solution of the optimization problem. This can be denoted symbolically as

$$\hat{H}_G \overset{g}{=} \hat{H}_{\mathrm{QUBO}} \tag{3.2}$$

where the right-hand side is the diagonal Hamiltonian corresponding to Eq. (3.1) and the left-hand side is the Rydberg-graph Hamiltonian realized by the actual atom array.

3.3 The Four Basic Gadgets

Four basic building blocks are needed to implement general QUBO.

(1) **Data qubit gadget**: implements a negative linear term $Q_{ii} < 0$. Mapping several atoms to the same variable x_i yields more Rydberg excitations—and thus lower energy—when that variable is 1.

(2) **Offset qubit gadget**: implements a positive linear term $Q_{ii} > 0$. Combining a data atom with an auxiliary atom offsets the energy gain at $x_i = 1$, producing a positive coefficient.

(3) **Even-atom quantum wire**: implements a positive quadratic term $Q_{ij} > 0$. Placing an even number of auxiliary atoms between two data qubits produces an energy cost when both variables are simultaneously 1.

(4) **Odd-atom quantum wire**: implements a negative quadratic term $Q_{ij} < 0$. Since a wire of an odd number of auxiliary atoms carries an additional linear term, it is used together with a data or offset gadget to implement a pure negative quadratic term.

Combining these, a general integer QUBO Hamiltonian can be expressed as

$$\hat{H}_{\text{QUBO}} \overset{g}{=} \sum_i \hat{H}_i^{\text{qubit}} + \sum_{i<j} \hat{H}_{ij}^{\text{wire}}. \tag{3.3}$$

3.4 Significance and Limitations

The advantage of this approach is that it implements the QUBO structure using only atom placement and auxiliary atoms, without per-atom local detuning or local laser-intensity control. This lowers the complexity of hardware control and allows diverse optimization problems to be addressed experimentally even under global control.

On the other hand, it has the drawback that the number of required auxiliary atoms grows as the coefficient magnitudes increase. In particular, QUBO problems with large weights can incur a large atom-number overhead. This method is therefore an important step demonstrating the universal implementability of QUBO, but its efficiency on actual large-scale problems is limited by the weight structure and the feasibility of hardware placement.

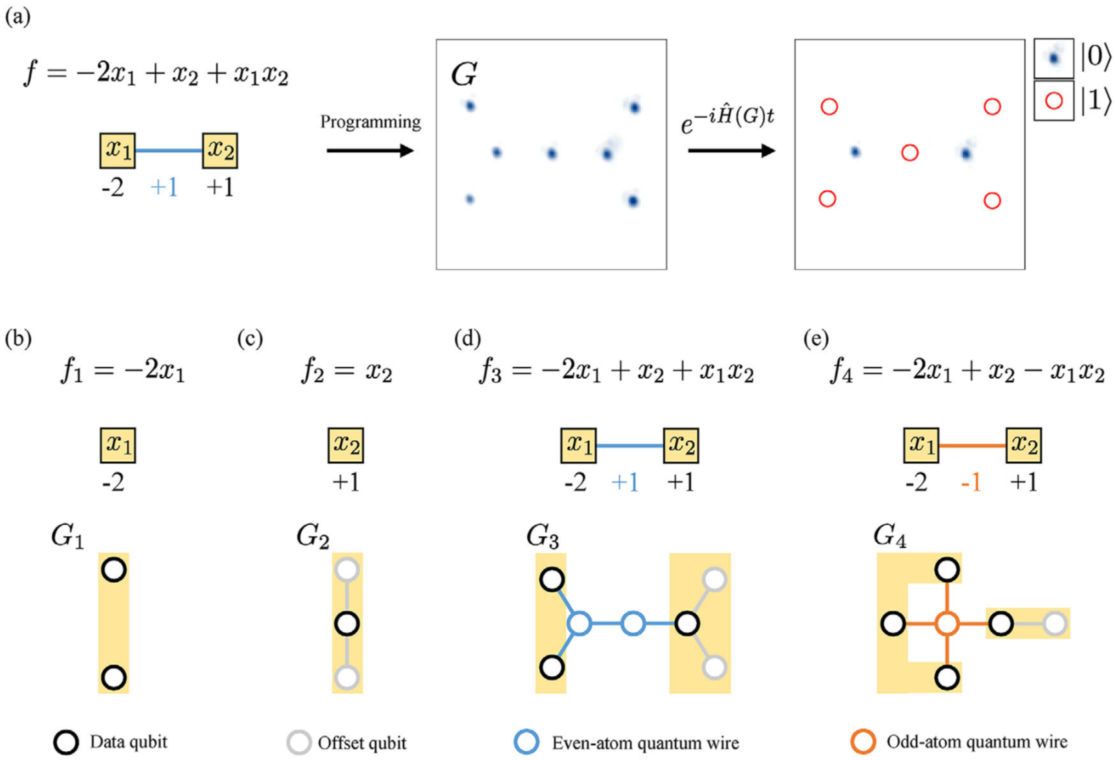

그림 3.1: Overview of the Rydberg-atom-graph-based QUBO implementation and the four basic gadgets. (a) Procedure for programming the weighted graph of the cost function $f = -2x_1 + x_2 + x_1x_2$ into a Rydberg-atom graph G. (b) data qubit gadget (G_1), (c) offset qubit gadget (G_2), (d) G_3 using an even-atom quantum wire, (e) G_4 using an odd-atom quantum wire. Source: Byun et al., *Adv. Quantum Technol.* (2024) [1].

제 4 장 Noisy Measurement and Deterministic Error Mitigation

Underlying study of this chapter: J. Park, J. Jung, and J. Ahn, "Quantum-Enhanced Deterministic Inference of k-Independent Set Instances on Neutral Atom Arrays," manuscript (2026) [2].

Author's contribution: The author (Junwoo Jung) made two key contributions to this study. First, the author derived the entropy-matching condition $H_2(p_{\text{eff}}) = f_0 H_2(p_{01}) + f_1 H_2(p_{10})$, which reduces the entropy cost of the DEM candidate space to a single effective error rate p_{eff} in the case of asymmetric measurement noise with $p_{01} \neq p_{10}$. Second, the author designed the Tarjan-DEM algorithm, which combines graph-theoretic pruning with exhaustive BF-DEM, and theoretically analyzed that its cost exponent is described by the same entropy constraint $c_0^{H_2(p)\cdot N}$. Experimental data acquisition and numerical simulation were carried out by lead author Park and co-authors.

4.1 Why Error Mitigation Is Needed

The output of a neutral-atom quantum annealing experiment is not the ideal ground-state distribution but the distribution of noisy measured bitstrings. Noise arises from various sources, including state-preparation-and-measurement errors, atom loss, control errors, and non-adiabatic transitions. In particular, even when the optimal solution is to some degree contained within the actual quantum state, flipping some bits during measurement can make the observed bitstring violate the independent-set condition or appear smaller than the target size.

Therefore, to fairly evaluate the performance of a quantum experiment, one must consider not merely the success rate of the raw measured bitstrings but also the cost of the classical postprocessing that infers a valid solution from noisy shots. From this perspective, the second study proposes and analyzes deterministic error mitigation (DEM).

4.2 Hamming-Shell Noise Model

Let the ideal MIS bitstring be $z_{\text{MIS}} \in \{0,1\}^N$. Assuming each bit independently flips with probability p, the Hamming distance between the measured bitstring z and the ideal bitstring is

$$HD(z, z_{\text{MIS}}) = \sum_{i=1}^{N} |z_i - z_{\text{MIS},i}| \tag{4.1}$$

which follows a binomial distribution with mean Np and variance $Np(1-p)$. That is, the noisy output is not spread uniformly over the full 2^N bitstring space but is concentrated in the Hamming shells around the ideal solution.

This structure turns error mitigation from a simple heuristic repair problem into an inference problem of searching for candidate solutions within a certain Hamming radius. If the noise rate is low, the candidate space to be searched is far smaller than the full space, and its size is governed by the binary entropy.

4.3 Deterministic Error Mitigation Algorithm

Deterministic error mitigation (DEM) generates candidate bitstrings around the measured bitstring z in order of Hamming distance and checks whether each candidate satisfies the independent-set condition and the target-size condition. In the decision version of the k-independent-set problem, one decides whether an independent set of size at least k exists.

Algorithm 1 Simplified deterministic error mitigation procedure

Input: graph $G = (V, E)$, target size k, measured bitstring z
for $i = 0, 1, 2, \ldots$ **do**
    for all w **such that** $HD(w, z) = i$ **do**
        if w is an independent set and $|w| \geq k$ **then**
            return w
        end if
    end for
end for

In the worst case this algorithm may search the full space 2^N, but under the noise model it on average needs to search only a Hamming ball of radius about Np. The candidate-space size is therefore approximated by

$$T(N, p) = \sum_{k=0}^{\lceil Np \rceil} \binom{N}{k} \approx \frac{2^{NH_2(p)}}{\sqrt{2\pi Np(1-p)}}, \tag{4.2}$$

where

$$H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p) \tag{4.3}$$

is the binary entropy. Equation (4.2) shows that the dominant cost scale of DEM is $2^{NH_2(p)}$.

4.4 Asymmetric Noise and the Effective Error Rate

In real measurements, $0 \to 1$ errors and $1 \to 0$ errors need not be symmetric. Denote these by $p_{01} = P(1|0)$ and $p_{10} = P(0|1)$. Letting the fractions of 0s and 1s in the ideal MIS be f_0 and f_1, asymmetric noise can be mapped to an effective symmetric error rate p_{eff} satisfying

$$H_2(p_{\text{eff}}) = f_0 H_2(p_{01}) + f_1 H_2(p_{10}). \tag{4.4}$$

This expression does not mean that the actual hardware noise can be reduced to a single bit-error rate; rather, it is an effective model for expressing the entropy cost of the DEM candidate space with the same dominant exponent.

4.5 Significance

The key significance of the DEM study is that it connects the output quality of a quantum experiment and the cost of classical postprocessing in the same computational language. It asks not simply "did the quantum device produce good samples?" but "how much classical computation is required to infer a valid solution from those samples?" This is a cost term that must be included when neutral-atom quantum optimization is compared with classical algorithms.

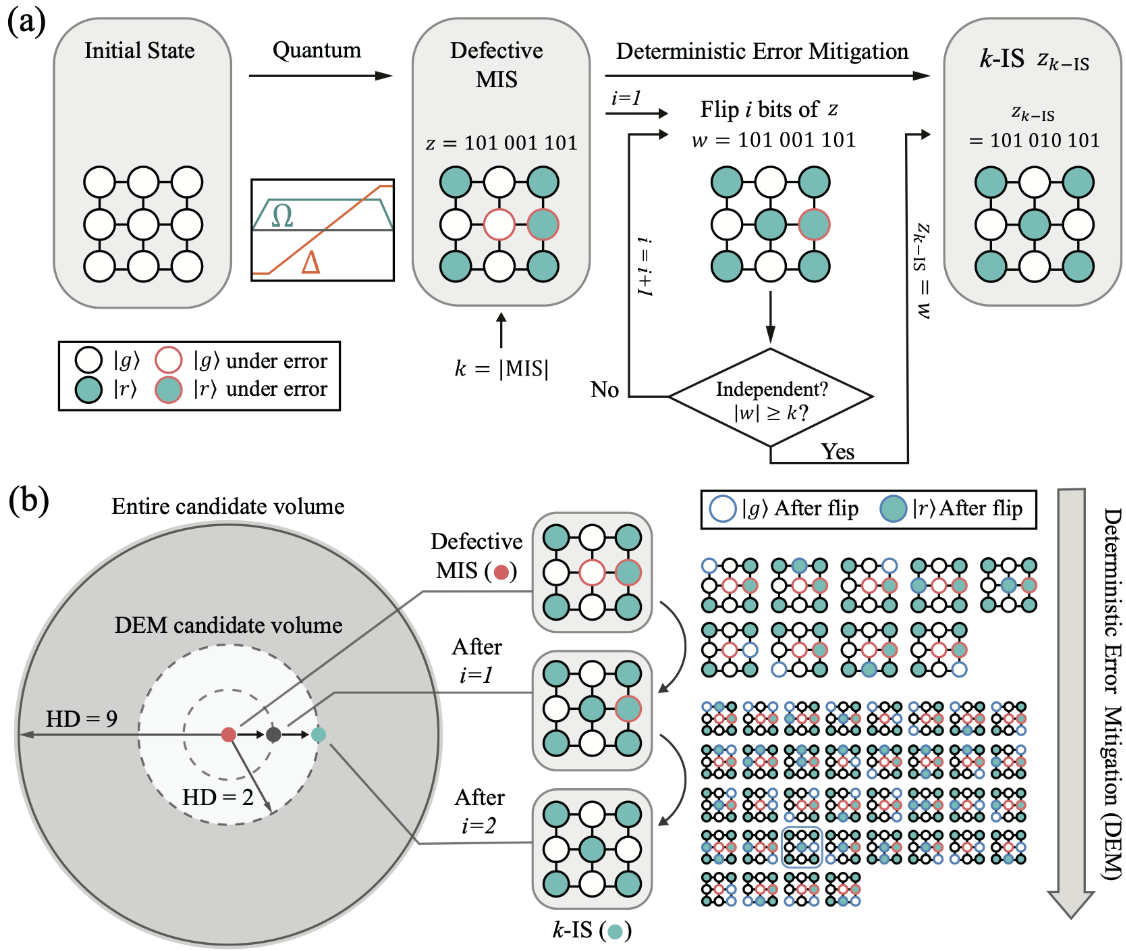

그림 4.1: Conceptual diagram of deterministic error mitigation (DEM). (a) The BF-DEM procedure, which generates candidate bitstrings in order of Hamming distance from the measured bitstring z of a Rydberg MIS experiment and checks the independent-set condition. (b) The geometric structure in which the DEM candidate space expands as the Hamming distance i increases. In the 3×3 lattice example with $N = 9$, $k = 5$, a valid k-independent set is found at $i = 2$. Source: Park et al., manuscript (2026) [2].

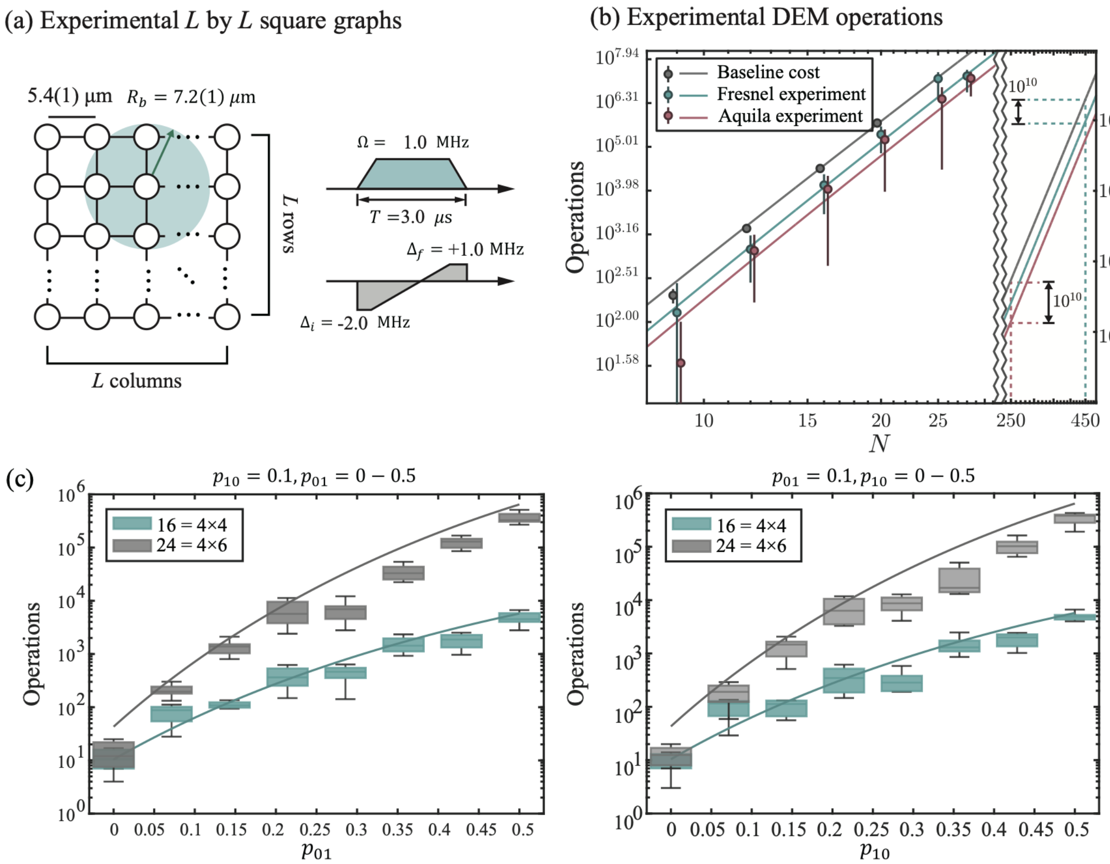

그림 4.2: Experimental results for the DEM postprocessing cost. (a) DEM operation count versus system size N measured on Pasqal Fresnel and QuEra Aquila devices. Colored markers are experimental data, gray markers the pure classical baseline, and solid lines the theoretical prediction $T(N, p_{\text{eff}})$ from the calibrated error rate p_{eff}. The exponential scaling governed by the binary entropy agrees well with experiment. (b,c) Comparison of numerical emulation under asymmetric noise $p_{01} \neq p_{10}$ with the effective-error-rate model. Source: Park et al., manuscript (2026) [2].

제 5 장 Approximate-Solution Shot Complexity and the Shell Model

Underlying study of this chapter: J. Jung and J. Ahn, "Shots-to-Approximate-Solution Scaling in Neutral-Atom Quantum Optimization," manuscript (2026) [3].

Author's contribution: This is a first-author study carried out by the author (Junwoo Jung) solely with Prof. Jaewook Ahn. The definition of the STS metric, the degeneracy-weighted shell model, the concept of the excitation-density-matched random baseline ($\Delta\beta_{\mathrm{ann}}$), the two-regime structure from a large-deviation perspective, the analysis of QuEra Aquila experimental data, and the writing of the entire manuscript were all carried out by the author.

5.1 Limitations of Exact-Solution-Centered Evaluation

Quantum optimization experiments are often evaluated by the number of shots required to obtain the exact optimum at least once. In actual optimization applications, however, a sufficiently good approximate solution may be more important than the exact optimum. In particular, for physically constrained unit-disk graphs or lattice graphs, the difference in difficulty between approximate and exact solutions can be central to performance evaluation.

The third study addresses this point by explicitly introducing a target approximation ratio r. If the independent-set size is $|S|$ and the MIS size is α, the approximation ratio is roughly $|S|/\alpha$. For a target approximation ratio r, success is defined as $|S| \geq \lceil r\alpha \rceil$.

5.2 Shell Index and Success Probability

Let the postprocessed output independent set be S_{post}. With MIS size $\alpha(G)$, the shell index is defined as

$$j = \alpha(G) - |S_{\mathrm{post}}|. \tag{5.1}$$

Here $j = 0$ means the exact MIS solution, and larger j is farther from the optimum. The target approximation ratio r defines the maximum allowed shell index

$$J(r) = \alpha - \lceil r\alpha \rceil. \tag{5.2}$$

The success probability of a single shot is therefore

$$p_r = \sum_{j=0}^{J(r)} \pi_j, \tag{5.3}$$

where π_j is the probability that a postprocessed shot belongs to shell j.

5.3 Approximate-Solution Shot-Count Metric

Let the target success confidence be p_{req}. The probability of succeeding at least once in n independent shots is $1-(1-p_r)^n$. The number of shots required to reach the target confidence is therefore defined as

$$\mathrm{STS}(r;N)=\left\lceil \frac{\ln(1-p_{\mathrm{req}})}{\ln(1-p_r)} \right\rceil . \tag{5.4}$$

This study uses $p_{\mathrm{req}}=0.99$.

The advantage of this metric is that it explicitly includes the approximation ratio r. With $r=1$ it demands the exact MIS, while $r<1$ permits a more relaxed target. The same experimental data can therefore require very different numbers of shots depending on the target quality.

5.4 Degeneracy-Weighted Shell Model

The postprocessed output distribution is well described by the following one-parameter model:

$$\pi_j(\beta)=\frac{d_{\alpha-j}e^{-\beta j}}{\sum_{u\geq 0} d_{\alpha-u}e^{-\beta u}}, \tag{5.5}$$

where $d_{\alpha-j}$ is the number of independent sets of size $\alpha-j$ and β is an effective inverse temperature indicating how concentrated the distribution is toward low shells, i.e., near the optimum. Larger β concentrates the distribution at small j.

This model expresses the competition of two effects. One is the degeneracy effect: shells slightly farther from the optimum may contain many more possible independent sets. The other is the effect by which quantum annealing and postprocessing concentrate the distribution toward low energy, i.e., low shells. Equation (5.5) expresses these two effects as the product of $d_{\alpha-j}$ and $e^{-\beta j}$.

5.5 Excitation-Density-Matched Random Baseline

That the postprocessed output appears high in quality cannot necessarily be attributed to quantum dynamics. If the mean excitation density p_{exc} of the raw bitstrings changes, the final independent-set size can change even when the same postprocessing algorithm is applied. To compare fairly with the quantum annealing output, a random baseline matched in excitation density is therefore needed.

To this end, Bernoulli random bitstrings with mean excitation density $p_{\mathrm{exc}}^{(\mathrm{ann})}$ are generated, and the same postprocessing as for the quantum data is applied. The resulting effective parameter is defined as $\beta_{\mathrm{rand}}(p_{\mathrm{exc}}^{(\mathrm{ann})})$. The residual concentration gain of quantum annealing can be expressed as

$$\Delta\beta_{\mathrm{ann}}=\beta_{\mathrm{ann}}-\beta_{\mathrm{rand}}(p_{\mathrm{exc}}^{(\mathrm{ann})}). \tag{5.6}$$

A positive value indicates that there exists low-shell concentration beyond the simple excitation-density effect.

5.6 Two Target-Dependent Regimes

An important conclusion of the STS analysis is that different regimes appear depending on the target approximation ratio. In the regime requiring the exact or near-exact solution ($r\approx 1$), the success

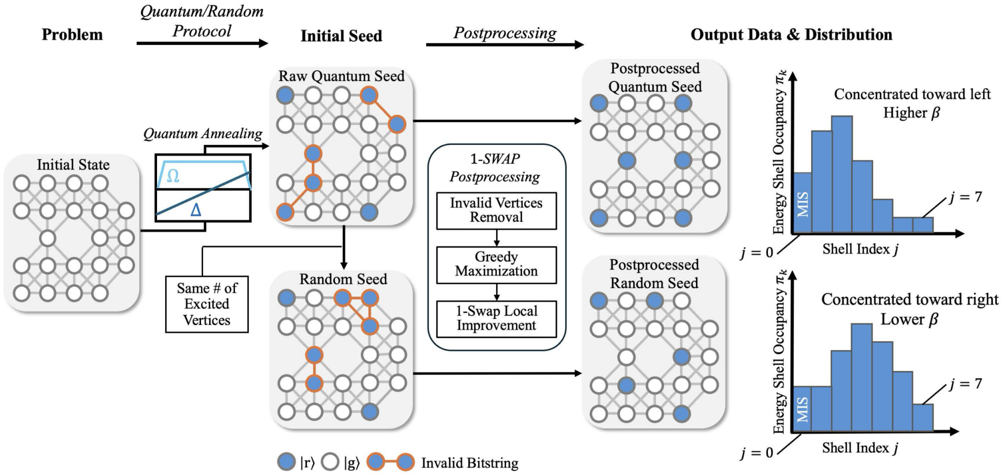

그림 5.1: Overall flow of the density-matched STS analysis. Raw bitstrings obtained by Rydberg quantum annealing (quantum seeds) and Bernoulli random bitstrings with the same excitation density (random seeds) are passed through the same postprocessing pipeline (independent-set projection → greedy expansion → 1-exchange improvement). The output independent sets are summarized by the shell index j, and the shell distribution is characterized via the effective inverse temperature β. A positive $\Delta\beta_{\text{ann}} > 0$ indicates additional concentration beyond the excitation-density effect. Source: Jung and Ahn, manuscript (2026) [3].

probability is determined by the probability mass of very small shells, so the required number of shots can grow exponentially with system size. In this case even a small improvement in β can make an exponential-scale difference.

Conversely, when r is low enough to permit a relaxed approximate solution, the range of accepted shells widens. The success probability then no longer corresponds to a rare tail probability but includes the typical region of the distribution, and the required number of shots can appear nearly constant. In this regime, even if quantum annealing produces a better distribution than the random baseline, the operationally required difference in shot count becomes small.

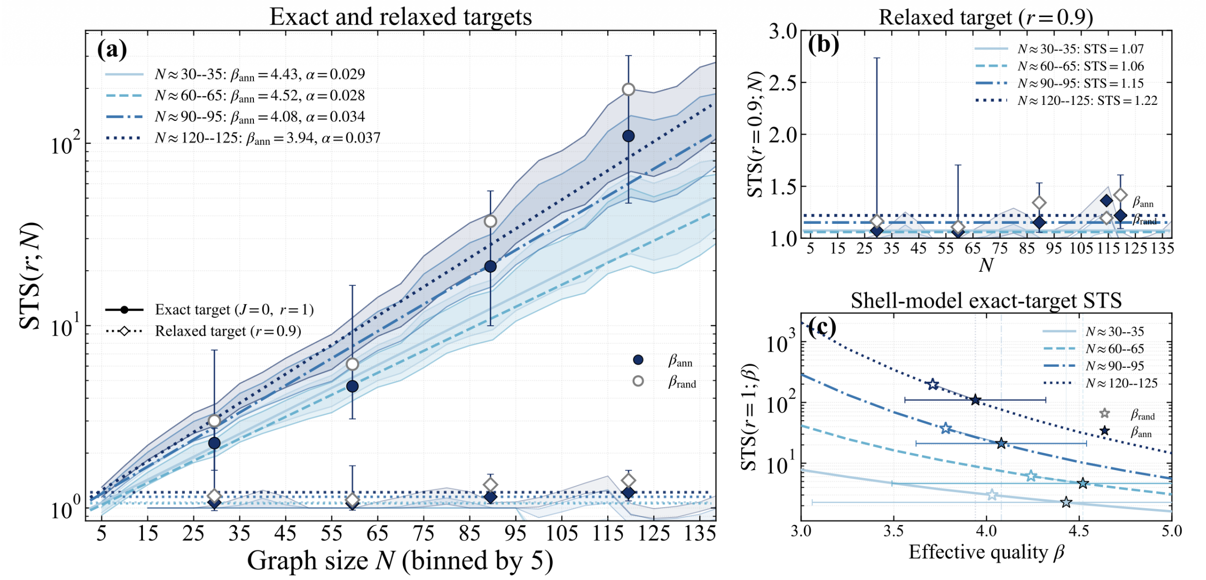

그림 5.2: Experimental results for the approximate-solution shot count STS. (a) System-size dependence of STS$(r; N)$ at the exact-solution target ($r = 1$). The four curves are exponential fits computed from the annealing plateau value $\beta_{\mathrm{ann}}^{(g)}$ of each size group, showing exponential growth with N. (b) At the relaxed target ($r = 0.9$), STS stays near 1 across the entire range of N. (c) STS$(r = 1; \beta)$ as a function of β. The annealing value ($\star$) lies to the right of the excitation-density baseline ($\bigstar$), showing a significant reduction of shot count in the exact-solution regime. Source: Jung and Ahn, manuscript (2026) [3].

제 6 장 Integrated Interpretation of the Three Studies

6.1 Three Levels: Encoding, Inference, and Evaluation

The three studies address different problems but connect naturally within the single pipeline of neutral-atom-based combinatorial optimization.

표 6.1: Roles and core questions of the three studies

Research topic	Position in pipeline	Core question
QUBO Rydberg graph	Problem encoding	Which cost functions can be implemented on an atom arr
DEM	Noise postprocessing & inference	How expensively can a valid solution be recovered from n
STS shell model	Performance evaluation	How does the shot count for a target approximation ratio

The first study addresses how to load the problem itself onto the hardware. The second addresses how to interpret the imperfect measurements the hardware produces. The third quantifies what the resulting output distribution means for actual optimization performance.

6.2 Shell Structure as a Common Language

Interestingly, the second and third studies both use shell structure as their core language. In DEM, the Hamming-distance shells between the measured bitstring and the ideal solution determine the size of the candidate space. In the STS study, the size difference j between the postprocessed independent set and the MIS determines the performance distribution. In both cases, instead of viewing the entire exponential space directly, one analyzes a structured subspace around the optimum.

This perspective fits well with the physical character of neutral-atom quantum optimization. The Rydberg blockade does not sample all bitstrings uniformly but samples a region restricted by the independent-set constraint and the energy structure. Noise, too, does not scatter the output uniformly across the full space but shifts it to the Hamming shells around the original state. Performance evaluation should therefore take place not over the full 2^N space but over the physically accessible shell structure.

6.3 The Difficulty of Defining Quantum Advantage

What the three studies show in common is that defining "quantum advantage" in quantum optimization is not straightforward. The fact that QUBO encoding is possible does not by itself guarantee a computational advantage. Even if a solution can be recovered from noisy shots via DEM, ignoring the cost of that classical postprocessing would not be a fair comparison. Moreover, even if the average solution quality appears good, without correcting for excitation density and postprocessing effects the pure contribution of the quantum dynamics cannot be known.

In this research workflow, therefore, quantum advantage must satisfy several conditions.

(1) The atom-number overhead of the problem encoding must be controllable.

(2) The classical cost of noise mitigation or postprocessing must be computed explicitly.

(3) When compared with a random or classical baseline, the simple density effect must be removed.

(4) Not only the exact solution but also the operational cost for each target approximation ratio must be presented.

These criteria show, without overstating the quantum device's performance, more honestly in which regimes it is actually useful.

제 7 장 Conclusion and Future Work

This graduation research report has organized combinatorial optimization on neutral-atom arrays into three levels: encoding, error mitigation, and performance evaluation. The Rydberg blockade naturally implements the MIS problem, and using auxiliary atoms and quantum wires, QUBO problems can be implemented even under global control. Since the actual experimental output is a noisy bitstring, however, valid solutions must be inferred through deterministic error mitigation based on the Hamming-shell structure. The postprocessing cost is then quantified by the candidate-space size governed by the binary entropy. Finally, the postprocessed output can be expressed as a distribution of shell indices from the MIS, and the number of shots required for a target approximation ratio can be evaluated by the STS metric.

Combining these three results leads to the conclusion that the performance of neutral-atom quantum optimization is difficult to summarize in a single number. The overhead of problem encoding, the measurement noise and postprocessing cost, and the success probability for a target approximation ratio must all be considered together. In particular, in the exact-solution regime quantum annealing may have an operational advantage through concentration toward low shells, but in the relaxed approximate-solution regime the required number of shots is already small, so the significance of any quantum advantage may diminish.

Future research directions can be organized into three. First, more efficient gadgets and placement strategies are needed to reduce the atom-number overhead of QUBO encoding. Second, beyond the simple exhaustive form of DEM, faster deterministic or probabilistic inference algorithms exploiting graph structure can be developed. Third, the STS analysis should be connected to actual time-to-solution and energy cost, extending it to a benchmarking metric that includes both quantum shot time and classical postprocessing time.

This work shows that neutral-atom arrays, in addressing combinatorial optimization problems, are not merely "devices that sample solutions" but hybrid computing platforms that combine physical encoding, noisy inference, and approximate-solution performance evaluation.

참 고 문 헌

[1] A. Byun, J. Jung, K. Kim, M. Kim, S. Jeong, H. Jeong, and J. Ahn, "Rydberg-Atom Graphs for Quadratic Unconstrained Binary Optimization Problems," *Advanced Quantum Technologies*, 2300398 (2024), doi:10.1002/qute.202300398.

[2] J. Park, J. Jung, and J. Ahn, "Quantum-Enhanced Deterministic Inference of k-Independent Set Instances on Neutral Atom Arrays," arXiv:2602.05432 (2026).

[3] J. Jung and J. Ahn, "Shots-to-Approximate-Solution Scaling in Neutral-Atom Quantum Optimization," manuscript (2026).

[4] M. Saffman, T. G. Walker, and K. Mølmer, "Quantum information with Rydberg atoms," *Reviews of Modern Physics* **82**, 2313–2363 (2010).

[5] A. Browaeys and T. Lahaye, "Many-body physics with individually controlled Rydberg atoms," *Nature Physics* **16**, 132–142 (2020).

[6] H. Pichler, S.-T. Wang, L. Zhou, S. Choi, and M. D. Lukin, "Quantum optimization for maximum independent set using Rydberg atom arrays," arXiv:1808.10816 (2018).

[7] S. Ebadi et al., "Quantum optimization of maximum independent set using Rydberg atom arrays," *Science* **376**, 1209–1215 (2022).

[8] J. Wurtz et al., "Aquila: QuEra's 256-qubit neutral-atom quantum computer," arXiv:2306.11727 (2023).

[9] M. Kim, K. Kim, J. Hwang, E.-G. Moon, and J. Ahn, "Rydberg quantum wires for maximum independent set problems," *Nature Physics* **18**, 755–759 (2022).

[10] A. Byun, M. Kim, and J. Ahn, "Finding the maximum independent sets of Platonic graphs using Rydberg atoms," *PRX Quantum* **3**, 030305 (2022).

[11] M.-T. Nguyen et al., "Quantum optimization with arbitrary connectivity using Rydberg atom arrays," *PRX Quantum* **4**, 010316 (2023).

[12] A. Lucas, "Ising formulations of many NP problems," *Frontiers in Physics* **2**, 5 (2014).

[13] T. Albash and D. A. Lidar, "Adiabatic quantum computation," *Reviews of Modern Physics* **90**, 015002 (2018).

[14] M. R. Garey and D. S. Johnson, *Computers and Intractability: A Guide to the Theory of NP-Completeness* (Freeman, New York, 1979).

[15] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. (Wiley, Hoboken, 2006).

[16] S. Jeong, J. Park, and J. Ahn, "Quantum-enhanced simulated annealing using Rydberg atoms," *Advanced Quantum Technologies*, e2500070 (2025).

[17] R. S. Andrist et al., “Hardness of the maximum independent set problem on unit-disk graphs and prospects for quantum speedups,” *Physical Review Research* **5**, 043277 (2023).

[18] E. T. Jaynes, “Information theory and statistical mechanics,” *Physical Review* **106**, 620–630 (1957).

사 사

My deep interest in neutral-atom quantum computing began when I first met Prof. Jaewook Ahn in Busan during my high school years. I later applied to his group in the summer of my first undergraduate year, and he kindly accepted me even though I did not yet know what I did not know. I am deeply grateful to Prof. Jaewook Ahn, who—when I had to learn even how to ask questions after beginning research—awakened in me the attitude and direction of thinking as a researcher. It was thanks to his insight and encouragement that I could complete this thesis.

I am also sincerely grateful to the senior colleagues with whom I worked in the lab. In particular, I thank Dr. Woojeong Byun, Dr. Kanghun Kim, Juyoung Park, and Kangjin Kim. Much of what I learned by watching and learning beside them remains throughout this thesis.

I am grateful as well to Prof. Jaeyoon Choi, Prof. Hawoong Jeong, and Prof. Gil Young Cho, who accepted me as an undergraduate intern during my studies, and to Prof. Nelly Ng of Nanyang Technological University in Singapore. I also extend deep thanks to Dr. Jungrak Son, Dr. Ray Ganardi, Dooyoung Cho, Deokyoung Lee, and Byungjoo Jang, who helped me greatly while we did research together.

Finally, I am most deeply grateful to my mother and father, who have always believed in and supported me unconditionally.